QP CODE Reg. No:……

Name:……

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION MARCH 2026

FOURTH SEMESTER

DSC A- MG4DSCBIF201-GENOMICS AND COMPUTATIONAL GENOMICS

(2024 ADMISSION ONWARDS)

Duration: 1.5 Hours Maximum Marks: 50

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability

Part A

Short Answer Questions

Answer any 5 Questions

Each Question carries 2 marks

1. Observe basic features of gene. [U]/[CO1]
2. Discuss RFLP. [U]/[CO2]
3. Discuss on Bac to Bac sequencing. [Ap]/[CO3]
4. Memorize nuclear genome. [K]/[C04]
5. Discover the role of Functional RNA's in human genome. [A]/[C05]
6. Describe regulatory elements. [Ap]/[C06]
7. Discover more on metagenomics. [A]/[C07]

[5x2=10]

Part B

Short Essay Type Questions

Answer any 4 Questions

Each Question carries 5 marks.

8. Interpret prokaryotic genome in deatil. [U]/[CO1]
9. Summarize genetic mapping. [U]/[CO2]
10. Identify the ethical issue related to sequencing ? [Ap]/[CO3]
11. Interpret nuclear and mitochondrial genome. [K]/ [CO4]
12. Discover types of RNA's . [A]/[CO5]
13. Examine the role of Genomic tool. [A][CO7]

[4x5 = 20]

Part C

Long Essay Type Questions

Answer any 2 Questions

Each Question carries 10 marks

14.Distinguish between prokaryotic and eukaryotic genome.	[U]/ [CO1]
15.Explain the outcome and applications of HGP	[Ap]/ [CO3]
16. Explain microarray and its types.	[Ap]/ [CO6]
17. Write a note on different genomic tools.	[A]/ [CO7]

[2x10 = 20]